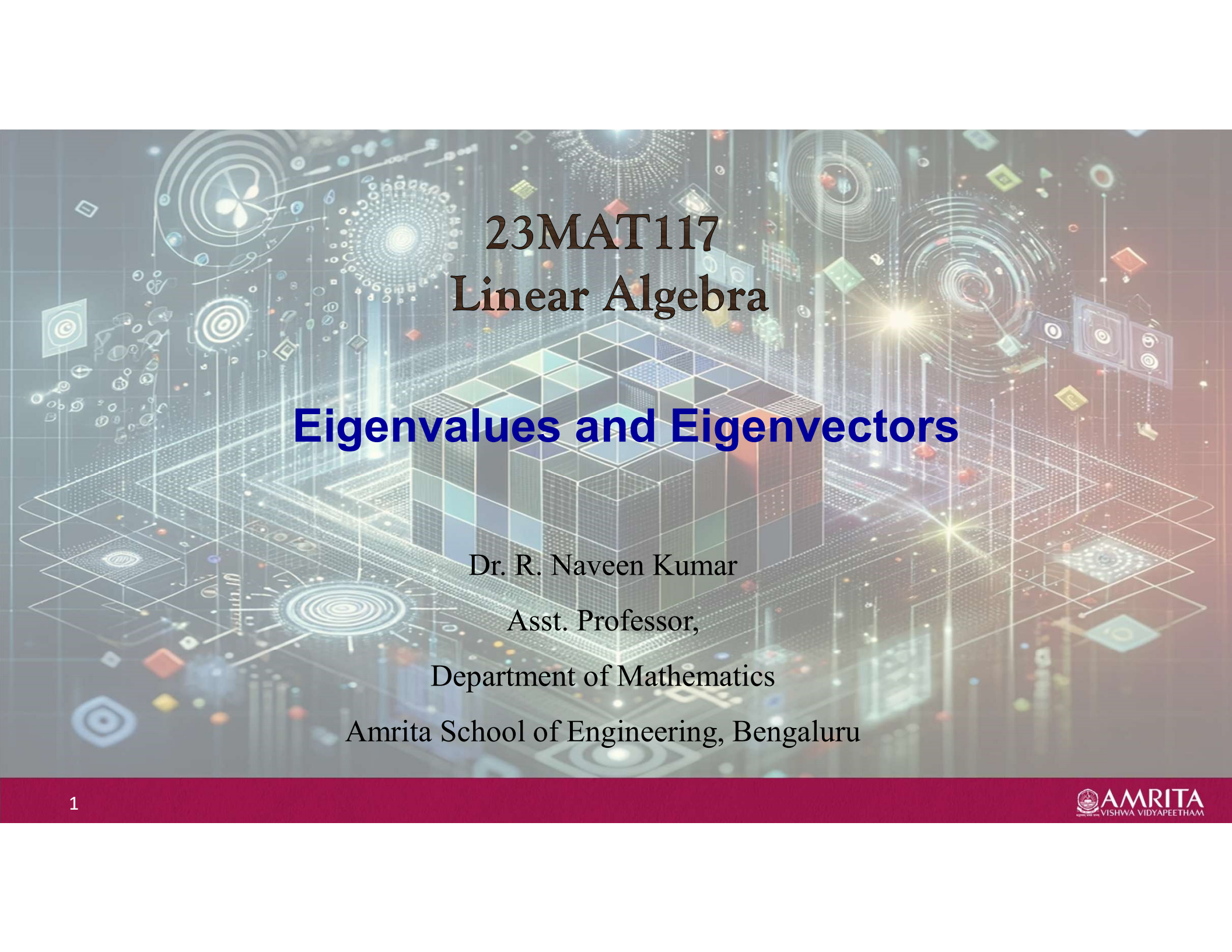




23MAT117 Linear Algebra

Eigenvalues and Eigenvectors

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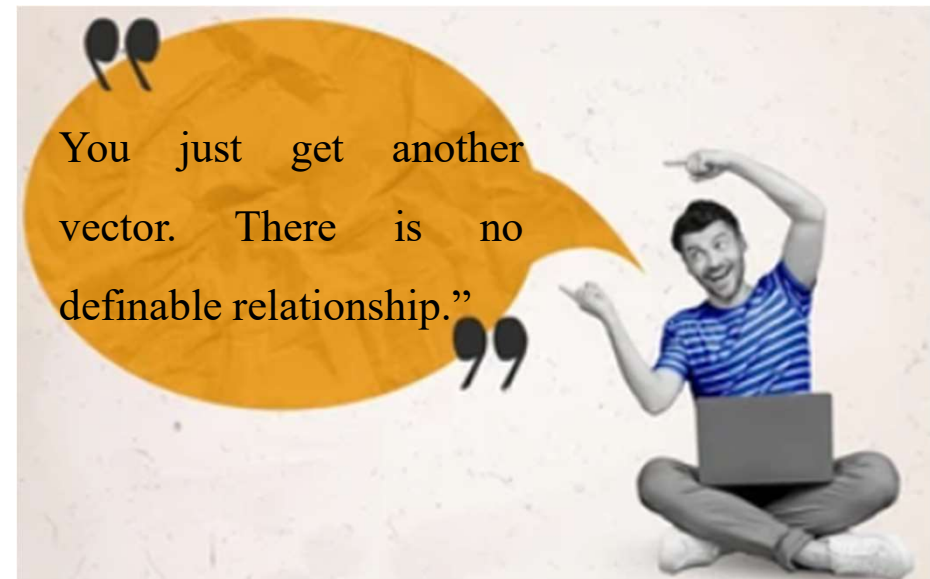
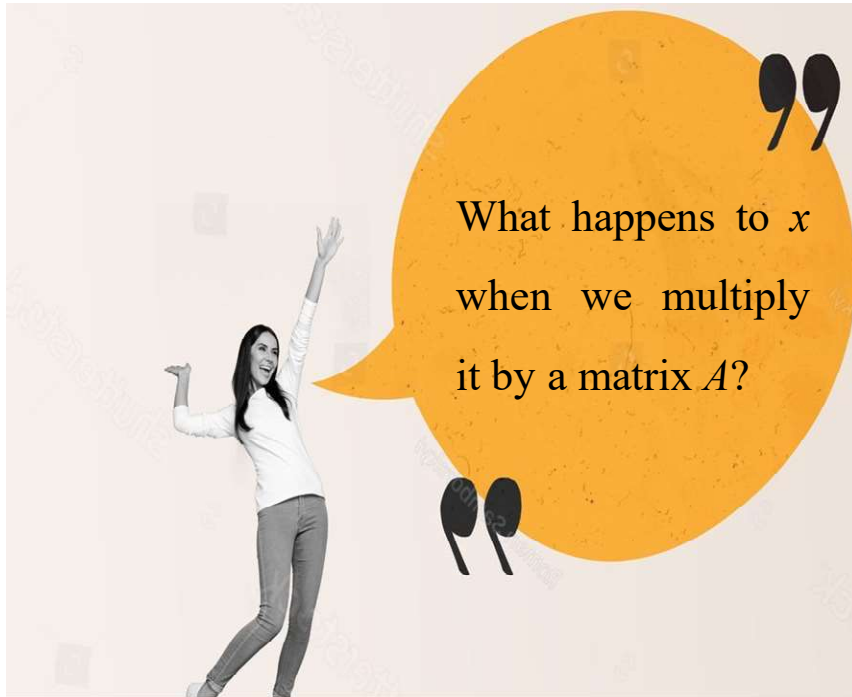
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What do we know about the result, when we multiply an unknown number x by another number, such as 2?



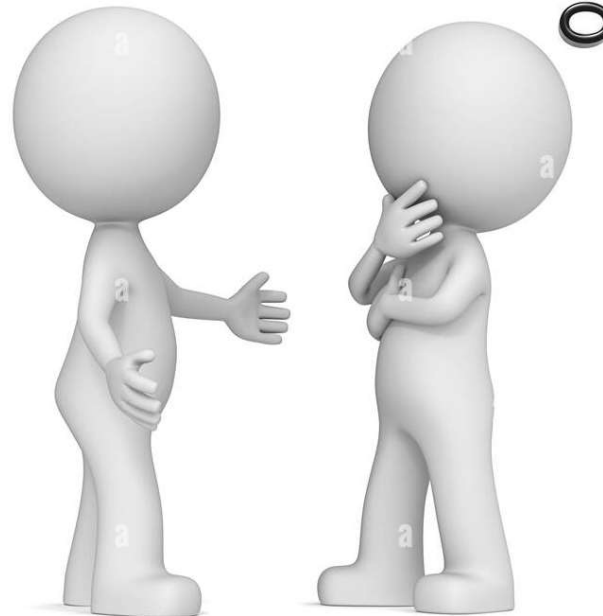
Unless, $x = 0$, we know that in some sense $2x$ will be “2 times bigger than x .”





Do we ever have the
case where $Ax = ax$,
where a is some
scalar?

???



Definition

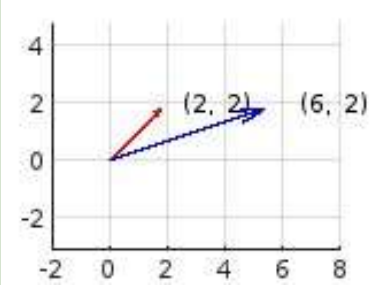
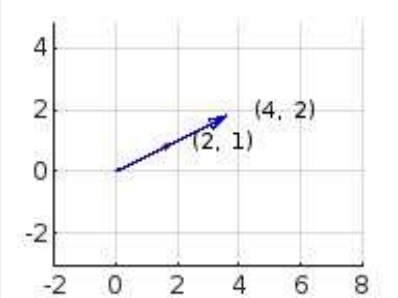
Let A be an $n \times n$ matrix, x a nonzero $n \times 1$ column vector, and λ a scalar.

If

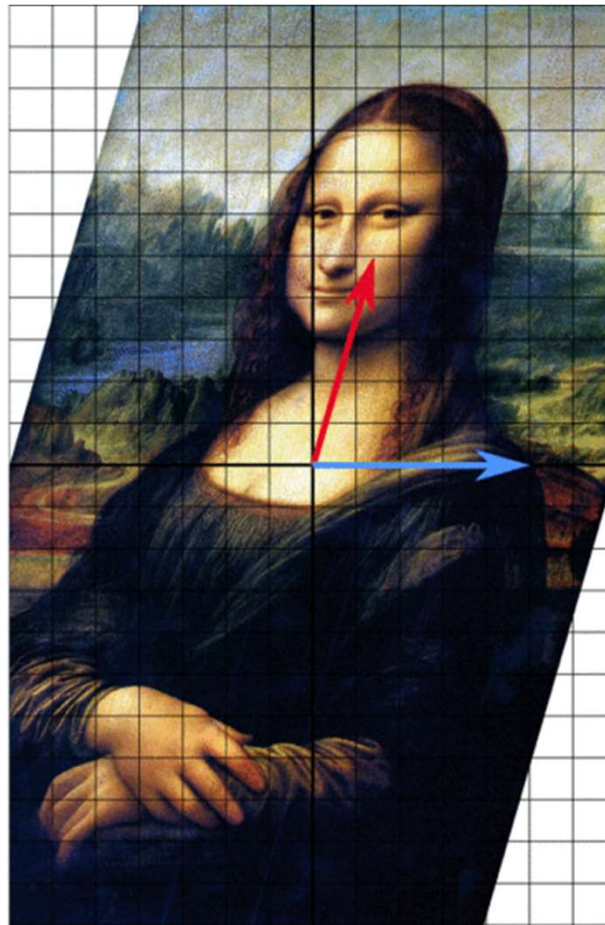
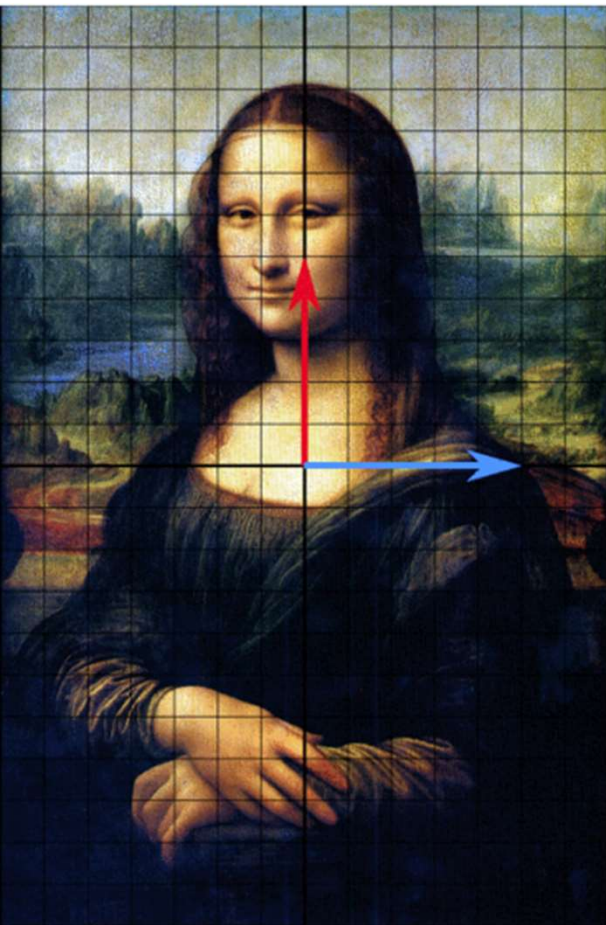
$$Ax = \lambda x,$$

then x is an eigenvector of A and λ is an eigenvalue of A .

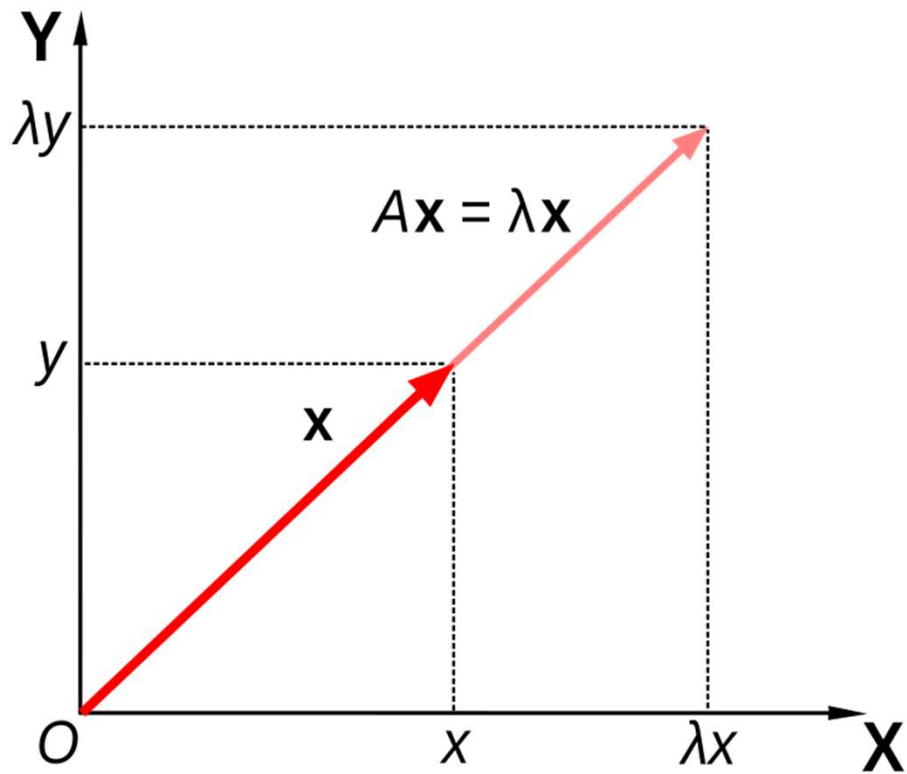
Eigen values and eigenvectors

A	x	Ax	Geometrical representation	Remarks	
$A = \begin{bmatrix} 1 & 2 \\ 1 & 0 \end{bmatrix}$	$\begin{bmatrix} 2 \\ 2 \end{bmatrix}$	$\begin{bmatrix} 6 \\ 2 \end{bmatrix}$		Scaled and rotated	
	$\begin{bmatrix} 2 \\ 1 \end{bmatrix}$	$\begin{bmatrix} 4 \\ 2 \end{bmatrix}$		Scaled without rotation	$Ax = \begin{bmatrix} 4 \\ 2 \end{bmatrix} = 2 \begin{bmatrix} 2 \\ 1 \end{bmatrix}$ $Ax = 2x$ $Ax = \lambda x$

Any vector obtained from scaling without rotation is called an eigenvector and the scaling factor is called an eigenvalue.



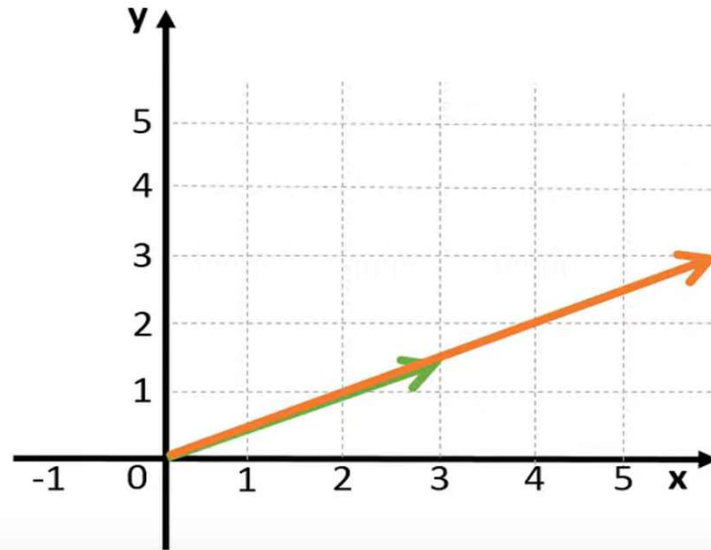
In this shear mapping the red arrow changes direction, but the blue arrow does not. The blue arrow is an eigenvector of this shear mapping because it does not change direction, and since its length is unchanged, its eigenvalue is 1.



Matrix A acts by stretching the vector $\mathbf{x}$, not changing its direction, so $\mathbf{x}$ is an eigenvector of A .

$$v = \begin{bmatrix} 3 \\ 1.5 \end{bmatrix}$$

$$A = \begin{bmatrix} 1 & 2 \\ 1 & 0 \end{bmatrix}$$



$$Av = \begin{bmatrix} 1 & 2 \\ 1 & 0 \end{bmatrix} \cdot \begin{bmatrix} 3 \\ 1.5 \end{bmatrix} = \begin{bmatrix} 6 \\ 3 \end{bmatrix} = 2 \begin{bmatrix} 3 \\ 1.5 \end{bmatrix}$$

Here the constant does not change

Consider the equation $AX = \lambda X \rightarrow \textcircled{1}$

If there is a scalar λ & a non-zero vector ' X ' such that $\textcircled{1}$ is satisfied. Then X is said to be an eigen vector corresponding to ' λ '.

$\rightarrow$ Eq $\textcircled{1}$ can also written as

$$(A - \lambda I)X = 0 \rightarrow \textcircled{2}$$

where I is the identity matrix of order $n \times n$.

Eq $\textcircled{2}$ represents the homogeneous linear system having ' n ' equations in ' n ' unknowns.

Since we want a non-zero vector ' X ' such that eq $\textcircled{2}$ is satisfied the rank of coefficient matrix should be less than ' n '.)

$\Rightarrow |A - \lambda I| = 0 \rightarrow \textcircled{3}$ This is called characteristic eq of ' A '

Solving eq $\textcircled{3}$ we get all the eigen values of ' A '. Substituting these eigen values in $\textcircled{2}$ one by one and solving the homogeneous system we get corresponding eigen vectors.

Finding eigenvalues and eigenvectors of a matrix

Given a square matrix A , we want to find a nonzero vector x and a scalar λ such that $Ax = \lambda x$.

$$Ax = \lambda x$$

$$\Rightarrow Ax - \lambda x = 0$$

$$\Rightarrow (A - \lambda I)x = 0$$

- This means that we want solutions to $(A - \lambda I)x = 0$ other than $x=0$.
- If the matrix $A - \lambda I$ is invertible, then the only solution is $(A - \lambda I)x = 0$ is $x = 0$.
- Therefore, in order to have other solutions, we need $A - \lambda I$ not to be invertible.
- The non invertible matrices all have determinant of 0.
- Therefore, if we want to find the eigen values λ and eigen vectors x , then $\det(A - \lambda I) = 0$.

Exploring the definition

Why A be a square matrix?
Why x must be nonzero?
What if $x = 0$?

If A isn't square, then Ax and λx will have different sizes, and so they cannot be equal.
If $x = 0$ then no matter what λ is, $Ax = \lambda x$ implies that every number is an eigenvalue; if every number is an eigenvalue, then we wouldn't need a definition for it!
Therefore we specify that $x \neq 0$.

Steps for finding eigenvalues and eigenvectors

- Calculate $A - \lambda I$.
- Then equate $\det(A - \lambda I) = 0$.
- The obtained equation is called as the characteristic equation.
- Find the roots of the characteristic equation to calculate the eigenvalues.
- Then with these eigen values, find the eigen vectors by substituting in the equation $(A - \lambda I)x = 0$

Example 1

Find the eigen values and eigen vectors of $A = \begin{bmatrix} 2 & -12 \\ 1 & -5 \end{bmatrix}$

Sol: $\det(A - \lambda I) = 0$

$$\Rightarrow \begin{vmatrix} 2-\lambda & -12 \\ 1 & -5-\lambda \end{vmatrix} = 0$$

$$\Rightarrow (2-\lambda)(-5-\lambda) + 12 = 0$$

$$\Rightarrow \lambda^2 + 3\lambda + 2 = 0 \quad \longrightarrow \quad \text{Characteristic equation}$$

$$\Rightarrow (\lambda + 1)(\lambda + 2) = 0$$

$$\Rightarrow \lambda = -1, -2$$

$$\lambda_1 = -1, \lambda_2 = -2 \quad \longrightarrow \quad \text{Eigen values}$$

For $\lambda_1 = -1$

$$\Rightarrow (A - \lambda_1 I)x = \begin{bmatrix} 3 & -12 \\ 1 & -4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 4t \\ t \end{bmatrix} = t \begin{bmatrix} 4 \\ 1 \end{bmatrix}, \quad t \neq 0$$

Eigenvector corresponding to the eigen value -1

For $\lambda_2 = -2$

$$\Rightarrow (A - \lambda_2 I)x = \begin{bmatrix} 4 & -12 \\ 1 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 3t \\ t \end{bmatrix} = t \begin{bmatrix} 3 \\ 1 \end{bmatrix}, \quad t \neq 0$$

Eigenvector corresponding to the eigen value -2

Ex. 2) Find the eigen values and eigen vectors of

$$\underline{\underline{A}} = \begin{bmatrix} -5 & 2 \\ 2 & -2 \end{bmatrix} \quad \begin{array}{l} -5x_1 + 2x_2 = \lambda x_1 \\ 2x_1 - 2x_2 = \lambda x_2 \end{array} \quad (\underline{\underline{A}} - \lambda \underline{\underline{I}})\underline{\underline{x}} = \underline{\underline{0}}$$

In homogeneous linear system, nontrivial solutions exist when $\det(\underline{\underline{A}} - \lambda \underline{\underline{I}}) = 0$.

Characteristic equation of **A**:

$$D(\lambda) = \det(\underline{\underline{A}} - \lambda \underline{\underline{I}}) = \begin{vmatrix} -5-\lambda & 2 \\ 2 & -2-\lambda \end{vmatrix} = \lambda^2 + 7\lambda + 6 = 0$$

Characteristic polynomial

↑
Characteristic determinant

Eigenvalues: $\lambda_1 = -1$ and $\lambda_2 = -6$

Eigenvectors: for $\lambda_1 = -1$, $\underline{\underline{x}}_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ for $\lambda_2 = -6$, $\underline{\underline{x}}_2 = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$

↑

obtained from Gauss elimination

EXAMPLE 1 Determination of Eigenvalues and Eigenvectors

We illustrate all the steps in terms of the matrix

$$A = \begin{bmatrix} -5 & 2 \\ 2 & -2 \end{bmatrix}.$$

Solution. (a) *Eigenvalues.* These must be determined *first*. Equation (1) is

$$Ax = \begin{bmatrix} -5 & 2 \\ 2 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \lambda \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}; \quad \text{in components,} \quad \begin{aligned} -5x_1 + 2x_2 &= \lambda x_1 \\ 2x_1 - 2x_2 &= \lambda x_2. \end{aligned}$$

Transferring the terms on the right to the left, we get

$$(2^*) \quad \begin{aligned} (-5 - \lambda)x_1 + 2x_2 &= 0 \\ 2x_1 + (-2 - \lambda)x_2 &= 0. \end{aligned}$$

This can be written in matrix notation

$$(3^*) \quad (A - \lambda I)x = 0$$

because (1) is $Ax - \lambda x = Ax - \lambda Ix = (A - \lambda I)x = 0$, which gives (3*). We see that this is a *homogeneous* linear system. By Cramer's theorem in Sec. 7.7 it has a nontrivial solution $x \neq 0$ (an eigenvector of A we are looking for) if and only if its coefficient determinant is zero, that is,

$$(4^*) \quad D(\lambda) = \det(A - \lambda I) = \begin{vmatrix} -5 - \lambda & 2 \\ 2 & -2 - \lambda \end{vmatrix} = (-5 - \lambda)(-2 - \lambda) - 4 = \lambda^2 + 7\lambda + 6 = 0.$$

We call $D(\lambda)$ the characteristic determinant or, if expanded, the characteristic polynomial, and $D(\lambda) = 0$ the characteristic equation of A . The solutions of this quadratic equation are $\lambda_1 = -1$ and $\lambda_2 = -6$. These are the eigenvalues of A .

(b₁) *Eigenvector of A corresponding to λ_1 .* This vector is obtained from (2*) with $\lambda = \lambda_1 = -1$, that is,

$$-4x_1 + 2x_2 = 0$$

$$2x_1 - x_2 = 0.$$

A solution is $x_2 = 2x_1$, as we see from either of the two equations, so that we need only one of them. This determines an eigenvector corresponding to $\lambda_1 = -1$ up to a scalar multiple. If we choose $x_1 = 1$, we obtain the eigenvector

$$\mathbf{x}_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \quad \text{Check:} \quad A\mathbf{x}_1 = \begin{bmatrix} -5 & 2 \\ 2 & -2 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} -1 \\ -2 \end{bmatrix} = (-1)\mathbf{x}_1 = \lambda_1 \mathbf{x}_1.$$

(b₂) *Eigenvector of A corresponding to λ_2 .* For $\lambda = \lambda_2 = -6$, equation (2*) becomes

$$x_1 + 2x_2 = 0$$

$$2x_1 + 4x_2 = 0.$$

A solution is $x_2 = -x_1/2$ with arbitrary x_1 . If we choose $x_1 = 2$, we get $x_2 = -1$. Thus an eigenvector of A corresponding to $\lambda_2 = -6$ is

$$\mathbf{x}_2 = \begin{bmatrix} 2 \\ -1 \end{bmatrix}, \quad \text{Check:} \quad A\mathbf{x}_2 = \begin{bmatrix} -5 & 2 \\ 2 & -2 \end{bmatrix} \begin{bmatrix} 2 \\ -1 \end{bmatrix} = \begin{bmatrix} -12 \\ 6 \end{bmatrix} = (-6)\mathbf{x}_2 = \lambda_2 \mathbf{x}_2.$$

For the matrix in the intuitive opening example at the start of Sec. 8.1, the characteristic equation is $\lambda^2 - 13\lambda + 30 = (\lambda - 10)(\lambda - 3) = 0$. The eigenvalues are $\{10, 3\}$. Corresponding eigenvectors are $[3 \ 4]^T$ and $[-1 \ 1]^T$, respectively. The reader may want to verify this. ■

Problems : (1) $A = \begin{pmatrix} 5 & 4 \\ 1 & 2 \end{pmatrix}$

$$1) A = \begin{pmatrix} 5 & 4 \\ 1 & 2 \end{pmatrix}$$

$$|A - \lambda I| = 0$$

$$\Rightarrow \begin{vmatrix} (5-\lambda) & 4 \\ 1 & (2-\lambda) \end{vmatrix} = 0$$

$$\Rightarrow \lambda^2 - 7\lambda + 10 - 4 = 0$$

$$\Rightarrow \lambda^2 - 7\lambda + 6 = 0$$

$$\Rightarrow \lambda^2 - 6\lambda - \lambda + 6 = 0$$

$$\Rightarrow \lambda(\lambda - 6) - 1(\lambda - 6) = 0$$

$$\Rightarrow \lambda - 1 \text{ or } 6$$

So we have eigen values of λ are 1 & 6.

We will now find eigen vectors corresponding to each eigen value.

For $\lambda = 1$, Consider the eqn.

$$[A - 1 \cdot I]X = 0$$

$$\Rightarrow \begin{bmatrix} 5-1 & 4 \\ 1 & 2-1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 4 & 4 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

The coefficient matrix $\begin{pmatrix} 4 & 4 \\ 1 & 1 \end{pmatrix}$

$$\text{is } \sim \begin{pmatrix} 4 & 4 \\ 0 & 0 \end{pmatrix} \quad \text{Perform } (R_2 \rightarrow R_2 - \frac{1}{4}R_1)$$

$$\Rightarrow \begin{pmatrix} 4 & 4 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = 0$$

$$\Rightarrow 4x_1 + 4x_2 = 0$$

$$\Rightarrow 4x_1 = -4x_2$$

$$x_1 = -x_2$$

$$\therefore \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} -x_2 \\ x_2 \end{pmatrix} = x_2 \begin{pmatrix} -1 \\ 1 \end{pmatrix}$$

So $\begin{pmatrix} -1 \\ 1 \end{pmatrix}$ is an eigen vector corresponding to the eigen value '1'.

For $\lambda = 6$, consider the eq $[A - 6I]X = 0$

For $\lambda = 6$, consider the eq $(A - \lambda I)x = 0$

$$\text{i.e.} \begin{bmatrix} 5-6 & 4 \\ 1 & 2-6 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} -1 & 4 \\ 1 & -4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

The coefficient matrix $\begin{bmatrix} -1 & 4 \\ 1 & -4 \end{bmatrix}$ is $\sim \begin{bmatrix} -1 & 4 \\ 0 & 0 \end{bmatrix}$ ($R_2 \rightarrow R_2 + R_1$)

$$\begin{pmatrix} -1 & 4 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = 0$$

$$-x_1 + 4x_2 = 0$$

$$x_1 = 4x_2$$

$$\therefore \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 4x_2 \\ x_2 \end{pmatrix} = x_2 \begin{pmatrix} 4 \\ 1 \end{pmatrix}$$

$\therefore \begin{pmatrix} 4 \\ 1 \end{pmatrix}$ is an e. vector corresponding to e. value '6'.

$$2) A = \begin{bmatrix} 4 & -6 & -6 \\ 0 & -2 & 0 \\ 1 & -1 & -1 \end{bmatrix}$$

The characteristic eqⁿ is

$$\begin{vmatrix} 4-\lambda & -6 & -6 \\ 0 & -2-\lambda & 0 \\ 1 & -1 & -1-\lambda \end{vmatrix} = 0.$$

$$\lambda^3 - \lambda^2 - 4\lambda + 4 = 0.$$

$$\lambda^2(\lambda-1) - 4(\lambda-1) = 0.$$

$$(\lambda-1)(\lambda^2-4) = 0.$$

$$\lambda = 1 \quad \text{or} \quad \lambda^2 - 4 = 0$$

$$\lambda^2 = 4$$

$$\lambda = \pm 2$$

$\therefore$ The eigen values of A are 1, 2, -2.

Now for $\lambda = 1$, consider the eqⁿ.

$$\begin{pmatrix} 4-1 & -6 & -6 \\ 0 & -2-1 & 0 \\ 1 & -1 & -1-1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} 3 & -6 & -6 \\ 0 & -3 & 0 \\ 1 & -1 & -2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

The coed matrix is $\left[\begin{array}{ccc|c} 3 & -6 & -6 & 0 \\ 0 & -3 & 0 & 0 \\ 1 & -1 & -2 & 0 \end{array} \right]$

$$2 \left[\begin{array}{ccc|c} 3 & -6 & -6 & 0 \\ 0 & -3 & 0 & 0 \\ 1 & -1 & -2 & 0 \end{array} \right]$$

$$R_1 \leftrightarrow R_3$$

$$2 \left[\begin{array}{ccc|c} 1 & -1 & -2 & 0 \\ 0 & -3 & 0 & 0 \\ 3 & -6 & -6 & 0 \end{array} \right]$$

$$R_3 \rightarrow R_3 - 3R_1$$

$$2 \left[\begin{array}{ccc|c} 1 & -1 & -2 & 0 \\ 0 & -3 & 0 & 0 \\ 0 & -3 & 0 & 0 \end{array} \right]$$

$$R_3 \rightarrow R_3 - R_2$$

$$\sim \begin{bmatrix} 1 & -1 & -2 \\ 0 & -3 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$\Rightarrow x_1 - x_2 - 2x_3 = 0$$

$$-3x_2 = 0$$

$$x_2 = 0$$

$$x_1 = 2x_3$$

$$\therefore \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 2x_3 \\ 0 \\ x_3 \end{pmatrix} = x_3 \begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix}$$

So $\begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix}$ is an e. vector corresponding to the e. value '1'.

$$\text{For } \lambda = 2, \text{ e. vector} = \begin{pmatrix} 3 \\ 0 \\ 1 \end{pmatrix}$$

$$\text{For } \lambda = -2, \text{ e. v. is } = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$$

Example

Find the eigenvalues and corresponding eigenvectors of $A = \begin{bmatrix} 1 & 2 & -1 \\ 1 & 0 & 1 \\ 4 & -4 & 5 \end{bmatrix}$

Sol:

The characteristic polynomial is

$$\begin{aligned}\lambda^3 - 6\lambda^2 + 11\lambda - 6 &= 0 \\ \Rightarrow (\lambda - 1)(\lambda - 2)(\lambda - 3) &= 0 \\ \Rightarrow \lambda &= 1, 2, 3\end{aligned}$$

Corresponding eigenvectors are

$$x_1 = \begin{bmatrix} -1 \\ 1 \\ 2 \end{bmatrix}, x_2 = \begin{bmatrix} -2 \\ 1 \\ 4 \end{bmatrix}, x_3 = \begin{bmatrix} -1 \\ 1 \\ 4 \end{bmatrix}$$

Extra problems

1. Find the eigen values and the corresponding eigen vectors of $\begin{bmatrix} 8 & -4 \\ 2 & 2 \end{bmatrix}$

$$\text{Characteristic matrix} = [A - \lambda I] = \begin{bmatrix} 8 - \lambda & -4 \\ 2 & 2 - \lambda \end{bmatrix}$$

Characteristic equation is $|A - \lambda I| = 0$

$$\Rightarrow \begin{vmatrix} 8 - \lambda & -4 \\ 2 & 2 - \lambda \end{vmatrix} = 0$$

$$(8 - \lambda)(2 - \lambda) + 8 = 0$$

$$\Rightarrow 16 + \lambda^2 - 10\lambda + 8 = 0$$

$$\Rightarrow \lambda^2 - 10\lambda + 24 = 0$$

$$\Rightarrow (\lambda - 6)(\lambda - 4) = 0$$

$$\Rightarrow \lambda = 6, 4 \text{ are eigen values of } A$$

$$\text{Consider the system } \begin{bmatrix} 8 - \lambda & -4 \\ 2 & 2 - \lambda \end{bmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = 0$$

Eigen vector corresponding to $\lambda = 4$

Put $\lambda = 4$ in the above system, we get $\begin{pmatrix} 4 & -4 \\ 2 & -2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$

$$\Rightarrow 4x_1 - 4x_2 = 0 \text{ --- (1)} \quad 2x_1 - 2x_2 = 0 \text{ --- (2)}$$

from (1) and (2) we have $x_1 = x_2$

Let $x_1 = \alpha$

$$\text{Eigen vector is } \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{bmatrix} \alpha \\ \alpha \end{bmatrix} = \alpha \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ is a Eigen vector of matrix A, corresponding eigen value $\lambda = 4$

Eigen Vector corresponding to $\lambda = 6$

put $\lambda = 6$ in the above system, we get $\begin{pmatrix} 2 & -4 \\ 2 & -4 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$

$$\Rightarrow 2x_1 - 4x_2 = 0 \text{ --- (1)} \quad 2x_1 - 4x_2 = 0 \text{ --- (2)}$$

from (1) and (2) we have $x_1 = 2x_2$

Let $x_2 = \alpha \Rightarrow x_1 = 2\alpha$

$$\text{Eigen vector} = \begin{bmatrix} 2\alpha \\ \alpha \end{bmatrix} = \alpha \begin{bmatrix} 2 \\ 1 \end{bmatrix} \quad \begin{bmatrix} 2 \\ 1 \end{bmatrix} \text{ is eigen vector of matrix A corresponding eigen value } \lambda = 6$$

2. Find the eigen values and the corresponding eigen vectors of matrix

$$\begin{bmatrix} 2 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 2 \end{bmatrix}$$

The characteristic equation is $|A - \lambda I| = 0$

The characteristic equation is $|A - \lambda I| = 0$

$$|A - \lambda I| = \begin{vmatrix} 2 - \lambda & 0 & 1 \\ 0 & 2 - \lambda & 0 \\ 1 & 0 & 2 - \lambda \end{vmatrix} = 0$$

$$\Rightarrow (2 - \lambda)(2 - \lambda)^2 - 0 + [-(2 - \lambda)] = 0$$

$$\Rightarrow (2 - \lambda)^3 - (\lambda - 2) = 0$$

$$\Rightarrow \lambda - 2 [-(\lambda - 2)^2 - 1] = 0$$

$$\Rightarrow \lambda - 2 [-\lambda^2 + 4\lambda - 3] = 0$$

$$\Rightarrow (\lambda - 2)(\lambda - 3)(\lambda - 1) = 0$$

$$\Rightarrow \lambda = 1, 2, 3$$

The eigen values of A is 1, 2, 3.

For finding eigen vector the system is $(A - \lambda I)X = 0$

$$\Rightarrow \begin{bmatrix} 2 - \lambda & 0 & 1 \\ 0 & 2 - \lambda & 0 \\ 1 & 0 & 2 - \lambda \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Eigen vector corresponding to $\lambda = 1$

$$\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$x_1 + x_3 = 0$$

$$x_2 = 0$$

$$x_1 + x_3 = 0$$

$$x_1 = -x_3, x_2 = 0$$

Let $x_3 = \alpha$

$$\Rightarrow x_1 = -\alpha, x_2 = 0, x_3 = \alpha$$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -\alpha \\ 0 \\ \alpha \end{bmatrix} = \alpha \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \text{ is Eigen vector}$$

Eigen vector corresponding to $\lambda = 2$

$$\begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Here $x_1 = 0$ and $x_3 = 0$ and we can take any arbitrary value x_2 i.e $x_2 = \alpha$ (say)

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ \alpha \\ 0 \end{bmatrix} = \alpha \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

Eigen vector is $\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$

Eigen vector corresponding to $\lambda = 3$

$$\begin{bmatrix} -1 & 0 & 1 \\ 0 & -1 & 0 \\ 1 & 0 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$-x_1 + x_3 = 0$$

$$-x_2 = 0$$

$$x_1 - x_3 = 0$$

here by solving we get $x_1 = x_3, x_2 = 0$ say $x_3 = \alpha$

$$x_1 = \alpha, \quad x_2 = 0, \quad x_3 = \alpha$$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} \alpha \\ 0 \\ \alpha \end{bmatrix} = \alpha \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

Eigen vector is $\begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$

3. Find the Eigen values and Eigen vectors of the matrix is $\begin{bmatrix} 3 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}$

Sol: Let $A = \begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}$

Consider characteristic equation is $|A - \lambda I| = 0$ Eigen Values $\lambda = 0, 3, 15$

Case (i): If $\lambda = 0$ $\begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

$$\Rightarrow 8x_1 - 6x_2 + 2x_3 = 0 \text{ --- (1)}$$

$$-6x_1 + 7x_2 - 4x_3 = 0 \text{ --- (2)}$$

$$2x_1 - 4x_2 + 3x_3 = 0 \text{ --- (3)}$$

Consider (2) & (3)

$$\begin{array}{ccc} x_1 & x_2 & x_3 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{array}$$

$$\Rightarrow \frac{x_1}{21-16} = \frac{-x_2}{-18+8} = \frac{x_3}{24-14} = k$$

$$\Rightarrow \frac{x_1}{5} = \frac{-x_2}{-10} = \frac{x_3}{10} = k$$

$$\Rightarrow x_1 = k, \quad x_2 = 2k, \quad x_3 = 2k$$

Eigen Vector is $\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} k \\ 2k \\ 2k \end{bmatrix} = k \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$

Case (ii): If $\lambda = 3$

$$\begin{bmatrix} 8-\lambda & -6 & 2 \\ -6 & 7-\lambda & -4 \\ 2 & -4 & 3-\lambda \end{bmatrix} x = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow 5x_1 - 6x_2 + 2x_3 = 0 \text{-----(1)}$$

$$-6x_1 + 4x_2 - 4x_3 = 0 \text{-----(2)}$$

$$2x_1 - 4x_2 + 0 = 0 \text{-----(3)}$$

$$\Rightarrow \frac{x_1}{0-16} = \frac{-x_2}{0+8} = \frac{x_3}{24-8} = k$$

$$\Rightarrow \frac{x_1}{-16} = \frac{-x_2}{8} = \frac{x_3}{16} = k$$

$$\Rightarrow \frac{x_1}{-2} = \frac{-x_2}{1} = \frac{x_3}{2} = k$$

$$\Rightarrow \frac{x_1}{-2} = k, \quad -x_2 = k, \quad x_3 = 2k$$

$$\Rightarrow x_1 = -2k, \quad x_2 = -k, \quad x_3 = 2k$$

$$\begin{bmatrix} 5 & -6 & 2 \\ -6 & 4 & -4 \\ 2 & -4 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Consider (2) & (3)

$$\begin{array}{ccc} x_1 & x_2 & x_3 \\ -6 & 4 & -4 \\ 2 & -4 & 0 \end{array}$$

Eigen Vector is $\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -2k \\ -k \\ 2k \end{bmatrix} = k \begin{bmatrix} -2 \\ -1 \\ 2 \end{bmatrix}$

Case (iii): If $\lambda = 15$

$$\begin{bmatrix} -7 & -6 & 2 \\ -6 & -8 & -4 \\ 2 & -4 & -12 \end{bmatrix} x = 0$$

$$\begin{bmatrix} -7 & -6 & 2 \\ -6 & -8 & -4 \\ 2 & -4 & -12 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow -7x_1 + (-6x_2) + 2x_3 = 0 \text{ ---- (1)}$$

$$-6x_1 - 8x_2 - 4x_3 = 0 \text{ ---- (2)}$$

$$2x_1 - 4x_2 - 12x_3 = 0 \text{ ---- (3)}$$

Consider (2) & (3)

$$\begin{array}{ccc} x_1 & x_2 & x_3 \\ -6 & -8 & -4 \\ 2 & -4 & -12 \end{array}$$

$$\Rightarrow \frac{x_1}{96-16} = \frac{-x_2}{72+8} = \frac{x_3}{24+16} = k$$

$$\Rightarrow \frac{x_1}{80} = \frac{-x_2}{80} = \frac{x_3}{40} = k$$

$$\Rightarrow \frac{x_1}{2} = k, \quad \frac{x_2}{2} = k, \quad \frac{x_3}{1} = k$$

$$\Rightarrow \hat{x}_1 = 2k, \quad \hat{x}_2 = 2k, \quad \hat{x}_3 = k$$

Eigen Vector is $\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 2k \\ -2k \\ k \end{bmatrix} = \begin{bmatrix} 2 \\ -2 \\ 1 \end{bmatrix} k$

4. Find the Eigen values and the corresponding Eigen vectors of the matrix.

$$\begin{bmatrix} -2 & 2 & -3 \\ 2 & 1 & -6 \\ -1 & -2 & -0 \end{bmatrix}$$

Sol: Let $A = \begin{bmatrix} -2 & 2 & -3 \\ 2 & 1 & -6 \\ -1 & -2 & -0 \end{bmatrix}$

The characteristic equation of A is $|A - \lambda I| = 0$ i.e. $\begin{vmatrix} -2-\lambda & 2 & -3 \\ 2 & 1-\lambda & -6 \\ -1 & -2 & -\lambda \end{vmatrix} = 0$

$$\Rightarrow (-2-\lambda)[- \lambda(1-\lambda)-12]-2[-2\lambda-6]-3[2(-2)+(1-\lambda)]=0$$

$$\Rightarrow \lambda^3 + \lambda^2 - 21\lambda - 45 = 0$$

$$\Rightarrow (\lambda + 3)(\lambda + 3)(\lambda - 5) = 0$$

$$\Rightarrow \lambda = -3, -3, 5$$

The Eigen values are -3,-3, and 5

Case (i): If $\lambda = -3$

$$\text{We get } \begin{bmatrix} -2+3 & 2 & -3 \\ 2 & 1+3 & -6 \\ -1 & -2 & 0+3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

The augment matrix of the system is
$$\left[\begin{array}{ccc|c} 1 & 2 & -3 & 0 \\ 2 & 4 & -6 & 0 \\ -1 & -2 & 3 & 0 \end{array} \right]$$

Performing $R_2 - 2R_1, R_3 + R_1$, we get
$$\left[\begin{array}{ccc|c} 1 & 2 & -3 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

Hence we have $x_1 + 2x_2 - 3x_3 = 0 \Rightarrow x_1 = -2x_2 + 3x_3$

Thus taking $x_2 = k_1$ and $x_3 = k_2$, we get $x_1 = -2k_1 + 3k_2; x_2 = k_1; x_3 = k_2$

Hence
$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = k_1 \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix} + k_2 \begin{bmatrix} 3 \\ 0 \\ 1 \end{bmatrix}$$

So $\begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}$ and $\begin{bmatrix} 3 \\ 0 \\ 1 \end{bmatrix}$ are the Eigen vectors corresponding to $\lambda = -3$

Case (ii): If $\lambda = 5$

We get
$$\begin{pmatrix} -7 & 2 & -3 \\ 2 & -4 & -6 \\ -1 & -2 & -5 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$\Rightarrow -7x_1 + 2x_2 - 3x_3 = 0 \text{ --- (1)}$$

$$2x_1 - 4x_2 - 6x_3 = 0 \text{ --- (2)}$$

$$-x_1 - 2x_2 - 5x_3 = 0 \text{ --- (3)}$$

Consider (2) & (3)

$$\begin{array}{ccc} x_1 & x_2 & x_3 \\ 2 & -4 & -6 \\ -1 & -2 & -5 \end{array}$$

$$\Rightarrow \frac{x_1}{20-12} = \frac{-x_2}{-10-6} = \frac{x_3}{-4-4} = k_3$$

$$\Rightarrow \frac{x_1}{8} = \frac{-x_2}{-16} = \frac{-x_3}{-8} = k_3$$

$$\Rightarrow \frac{x_1}{1} = \frac{-x_2}{-2} = \frac{-x_3}{-1} = k_3$$

Eigen vector is
$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ -2 \\ -1 \end{bmatrix} k_3$$

Properties Of Eigen Values

1) A square matrix of order ' n ' has ' n ' eigen values counting multiplicity. These ' n ' eigen need not be distinct.

In any case they are going to be real no's (or) complex conjugate.

2) The sum of eigen values of a matrix is equal to trace of that matrix.

3) The determinant of a matrix is equal to the product of eigen values.

- 4) If the matrix is singular, then '0' will be one of eigen value for that matrix.
- 5) A matrix is invertible if and only if, all the eigen values are non-zero.
- 6) If $\lambda_1, \lambda_2, \dots, \lambda_n$ are eigen values of a matrix 'A', $k\lambda_1, k\lambda_2, \dots, k\lambda_n$ are eigen value of KA , where k is a non-zero scalar.
- 7) If $\lambda_1, \lambda_2, \dots, \lambda_n$ are eigen values of a matrix 'A' then any for +ve integer 'm', $\lambda_1^m, \lambda_2^m, \dots, \lambda_n^m$ are the eigen values of A^m .

8) If $\lambda_1, \dots, \lambda_n$ are eigen values of 'A' and if 'A' is invertible then $\frac{1}{\lambda_1}, \frac{1}{\lambda_2}, \dots, \frac{1}{\lambda_n}$ will be eigen values of A^{-1} .

$$\text{Let } AX = \lambda_1 X.$$

$$(A^{-1}A)X = IX = X.$$

$$A^{-1}(AX) = X \Rightarrow A^{-1}(\lambda_1 X) = X.$$

$$A^{-1}X = \frac{1}{\lambda_1} X.$$

9) For a diagonal matrix the eigen values are precisely the entries in the main diagonal. Also triangular matrix, the diagonal elements are precisely the eigen values.

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix}$$

SPECTRUM:-

The set of all eigen values of a matrix 'A' called spectrum of A.

Algebraic Multiplicity:-

The algebraic multiplicity of an eigen value λ is the no. of times that particular eigen value occurs in the spectrum of 'A'.

For Eg

If $A = \begin{pmatrix} 1 & 0 & 0 \\ 4 & 1 & 0 \\ 3 & 6 & 2 \end{pmatrix}$ then eigen values are 1 & 2.

The algebraic multiplicity of '1' is '2'.

The algebraic multiplicity of '2' is '1'.

Spectrum of $A = \{1, 1, 2\}$.

Geometric Multiplicity:-

The geometric Multiplicity of an eigen value λ is the no. of linearly independent eigen vectors corresponding to λ .

When $A = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$,

spectrum $A = \{1, 1\}$.

Here $(0,1)$ & $(1,0)$ are the two L.I. e-vectors corresponding to the e-value 1.

$\therefore$ Geometric multiplicity of e-value 1 is 2.

Theorem:-

Alg. Multiplicity of λ - Geometric Mult: of λ is called the

Defect of A .

Defect is always ≥ 0 .

Algebraic Mult of $\lambda \geq$ Geo. Multiplicity of λ .

1. Find the e-values of a 2×2 singular matrix having trace = 10.

2. Let $A = \begin{bmatrix} 100 & 100 \\ 100 & 100 \end{bmatrix}$

Find the spectrum of A .

3) Suppose A is a 7×7 singular matrix and $1-2i, 1+6i, 1-3i$ are in spectrum of A . Find all the element in spectrum of A .

4) What is order of matrix A if its characteristic polynomial is $\lambda^9 - 7\lambda^8 - 6\lambda$? Is A invertible?

5) If the e-values of a 3×3 matrix A are $2, 3, -1$ find the e-values of $A^2, 3A, A^{-1}, 9A^{-1}$.

Answers

1) Let λ_1 & λ_2 be the e-value of A .

W-k-T $\lambda_1 + \lambda_2 = \text{tr}(A)$.

$\lambda_1 \cdot \lambda_2 = \det(A)$.

It is given that $\det(A) = 0$
& $\text{tr}(A) = 10$

$\det(A) = 0 \Rightarrow$ either $\lambda_1 = 0$ or $\lambda_2 = 0$.

Assume without loss of generality that $\lambda_1 = 0$.

Since $\lambda_1 + \lambda_2 = 10$

$\lambda_1 = 0 \Rightarrow \lambda_2 = 10$

So $\text{Spec}(A) = \{0, 10\}$

2) Here $A = \begin{bmatrix} 100 & 100 \\ 100 & 100 \end{bmatrix}$.

A is a singular matrix.

$\det(A) = 0$.

$\lambda_1 \lambda_2 = 0$.

$\lambda_1 = 0$ (or) $\lambda_2 = 0$

$\text{tr}(A) = 200$

$\lambda_1 + \lambda_2 = 200$

Let $\lambda_1 = 0 \Rightarrow \lambda_2 = 200$.

$\therefore \text{Spec}(A) = \{0, 200\}$.

3) $\text{Spec } A = \{1-2i, \underline{1+2i}, 1+6i, \underline{1-6i}, 1-3i, \underline{1+3i}, 0\}$

$$4) \lambda^9 - 7\lambda^8 - 6\lambda = 0$$

$$\lambda(\lambda^8 - 7\lambda^7 - 6) = 0$$

$$\lambda = 0 \quad \text{or} \quad \lambda^8 - 7\lambda^7 - 6 = 0.$$

Here one of eigen value is zero then, its det is '0'

$\therefore A$ is not invertible. $\because \text{order} = 9$

$$5) \text{spec}(A^2) = \{4, 9, 1\}$$

$$\text{spec}(3A) = \{6, 9, -3\}$$

$$\text{spec}(A^{-1}) = \{\frac{1}{4}, \frac{1}{9}, -1\}$$

$$\text{spec}(9A) = \{\frac{9}{2}, 3, 1\}$$

Exercise problems

1. Find the eigen values of the matrix $A = \begin{bmatrix} 1 & 2 & 1 \\ 0 & -5 & 0 \\ 1 & 8 & 11 \end{bmatrix}$.
2. Let A be a 3×3 matrix with real entries such that $\det(A) = 6$ and the trace of A is 0. If the $\det(A + I) = 0$, where I denote a 3×3 identity matrix, then find the eigen values of A .
3. Let A be a 3×3 real matrix with $\det(A + iI) = 0$, where $i = \sqrt{-1}$ and I is the 3×3 identity matrix. Find the trace of A^2 if $\det(A) = 3$.

Question 2

Given $\det(A) = 6$, $\text{tr}(A) = 0$ and $\det(A+I) = 0$

Let the eigen values of A be λ_1, λ_2 and λ_3 .

* Since $\det(A) = 6$

$$\Rightarrow \lambda_1 \cdot \lambda_2 \cdot \lambda_3 = 6 \rightarrow (1)$$

* Since $\text{tr}(A) = 0$

$$\Rightarrow \lambda_1 + \lambda_2 + \lambda_3 = 0 \rightarrow (2)$$

* $\det(A+I) = 0$

$$\Rightarrow (\lambda_1 + 1)(\lambda_2 + 1)(\lambda_3 + 1) = 0$$

$\Rightarrow$ atleast one of $\lambda_i + 1$ is zero.

$$\Rightarrow \lambda_i + 1 = 0$$

$$\Rightarrow \lambda_i = -1$$

Suppose $\lambda_1 = -1$.

Then (1) $\Rightarrow -\lambda_2 \lambda_3 = 6$

and (2) $\Rightarrow -1 + \lambda_2 + \lambda_3 = 0$

Suppose $\lambda_1 = -1$.

Then (1) $\Rightarrow -\lambda_2 \lambda_3 = 6$

and (2) $\Rightarrow -1 + \lambda_2 + \lambda_3 = 0$

$$\lambda_2 \lambda_3 = -6$$

$$\lambda_2 + \lambda_3 = 1$$

$$-\frac{6}{\lambda_3} + \lambda_3 = 1$$

$$\lambda_3^2 - \lambda_3 - 6 = 0$$

$$\lambda_3^2 - 3\lambda_3 + 2\lambda_3 - 6 = 0$$

$$\lambda_3 [\lambda_3 - 3] + 2 [\lambda_3 - 3] = 0$$

$$\Rightarrow \lambda_3 = -2, 3$$

When $\lambda_3 = -2$;

$$\lambda_2 = \frac{-6}{-2} = 3 //$$

When $\lambda_3 = 3$

$$\lambda_2 = -6/3 = -2 //$$

-2, 3 and 3, -2

↓

$$\lambda_1 = -1$$

↓

$$\lambda_1 = -1$$

$\therefore$ The eigen values are -1, -2 and 3

Question 3

Given $\det(A + iI) = 0 \Rightarrow -i$ is an eigen value of A .

Since complex number comes in pair, i is also an eigen value of A .

We have $\det(A) = 3$.

Let the third eigen value be x .

$$\text{Then } i \times -i \times x = 3$$

$$\Rightarrow \underline{\underline{x = 3}}$$

So the eigen values are i , $-i$ and 3 .

Then the eigen values of A^2 are $(i)^2$, $(-i)^2$ and 3^2
 $\Rightarrow -1, -1$ and 9

The trace of A^2 = Sum of eigen values of A^2

$$= -1 + -1 + 9$$

$$= \underline{\underline{7}}$$

PROBLEM SET 8.1

1–16 EIGENVALUES, EIGENVECTORS

Find the eigenvalues. Find the corresponding eigenvectors. Use the given λ or factor in Probs. 11 and 15.

1. $\begin{bmatrix} 3.0 & 0 \\ 0 & -0.6 \end{bmatrix}$

2. $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$

3. $\begin{bmatrix} 5 & -2 \\ 9 & -6 \end{bmatrix}$

4. $\begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$

5. $\begin{bmatrix} 0 & 3 \\ -3 & 0 \end{bmatrix}$

6. $\begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix}$

7. $\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$

8. $\begin{bmatrix} a & b \\ -b & a \end{bmatrix}$

9. $\begin{bmatrix} 0.8 & -0.6 \\ 0.6 & 0.8 \end{bmatrix}$

10. $\begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$

11. $\begin{bmatrix} 6 & 2 & -2 \\ 2 & 5 & 0 \\ -2 & 0 & 7 \end{bmatrix}, \lambda = 3$

12. $\begin{bmatrix} 3 & 5 & 3 \\ 0 & 4 & 6 \\ 0 & 0 & 1 \end{bmatrix}$

13. $\begin{bmatrix} 13 & 5 & 2 \\ 2 & 7 & -8 \\ 5 & 4 & 7 \end{bmatrix}$

14. $\begin{bmatrix} 2 & 0 & -1 \\ 0 & \frac{1}{2} & 0 \\ 1 & 0 & 4 \end{bmatrix}$

15. $\begin{bmatrix} 1 & 0 & 0 \\ 2 & 4 & 0 \\ 6 & 4 & 2 \end{bmatrix}$

16. $\begin{bmatrix} 0.5 & 0.2 & 0.1 \\ 0 & 1.0 & 1.5 \\ 0 & 0 & 3.5 \end{bmatrix}$

17. $\begin{bmatrix} 4 & -2 & 3 \\ -2 & 1 & 6 \\ 1 & 2 & 2 \end{bmatrix}, \lambda = -3$

18. $\begin{bmatrix} 3 & 0 & 12 \\ -6 & 3 & 0 \\ 9 & 6 & 3 \end{bmatrix}, \lambda = 9$

19. $\begin{bmatrix} 13 & 5 & 2 \\ 2 & 7 & -8 \\ 5 & 4 & 7 \end{bmatrix}$

